

Learned-covariance sequential Kalman filter

two networks, one differentiable Kalman layer

The idea

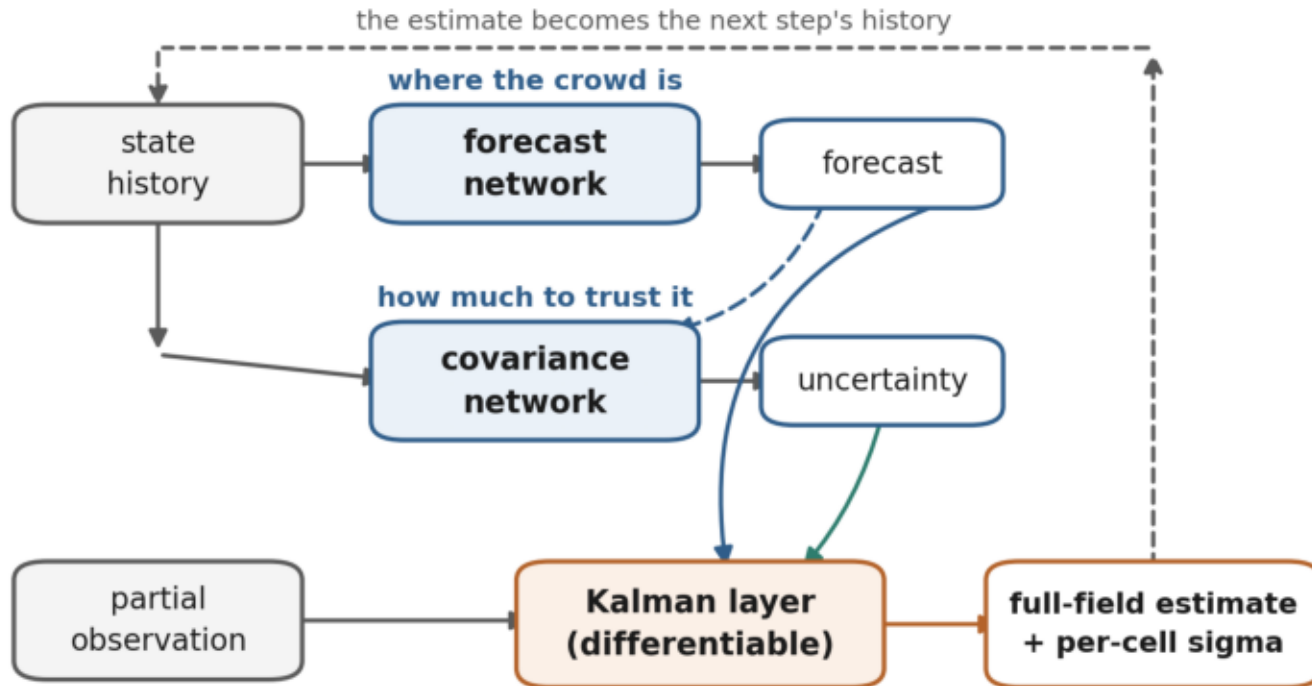
A filter needs two things: a forecast, and a sense of how far to trust it.

We learn both. One network predicts where the crowd will be. A second predicts how uncertain that prediction is, cell by cell.

A differentiable Kalman layer fuses the two with whatever the robots happened to see, and returns a full-field estimate together with a per-cell uncertainty.

Because the layer is differentiable, the whole thing trains end to end - which matters, because nobody can label a covariance directly.

The framework



The robots see only part of the corridor.

The forecast covers all of it; the learned uncertainty decides, cell by cell, whether to move towards what was measured or to stay with what was predicted.

It runs sequentially: each estimate becomes part of the next step's history.